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Problem Set 5 Corrections:

Q1 c) had some typos and Q1 d) is not bounded.

(c) Let $A_k = \{(x_1, \ldots, x_k) \in R^k | x_i = \frac{1}{k}, \forall i \leq k\}$. Suppose for some $n \geq 1$, $A = \cup_{k=1}^{n} A_k$, $k \leq n$. Which of the 4 properties mentioned above does A satisfy? Ans. $A = \{1, (\frac{1}{2}, \frac{1}{2}), \ldots, (\frac{1}{n}, \ldots, \frac{1}{n})\}$. It is Closed andBounded.

(d) $D = A \setminus B$ where $A = \{(x_1, \ldots, x_n) \in R^n | x_i \leq 5\}$ and $B = \{(x_1, \ldots, x_n) \in R^n | 4 < x_i \leq 5\}$. Ans. The set D is simply $\{(x_1, \ldots, x_n) \in R^n | x_i \leq 4, \forall i = 1, 2 \ldots, n\}$. It is convex and closed, but not bounded.

There might be some errors in some questions throughout the course.